

CHAPTER 4

Principles of Inheritance and Variation

Mendel's Laws of Inheritance And Inheritance of One Gene

1. The mother has A⁺ blood group, the father has B⁺ and the child is A⁺. What can be the possible genotypes of all three, respectively? (2024 Re)

A. $I^A I^A | I^B i | I^B i$ B. $I^A I^A | I^B i | I^A i$
C. $I^B i | I^A I^A | I^A I^B$ D. $I^A I^A | I^B I^B | I^A i$
E. $I^A i | I^B i | I^A i$

Choose the correct answer from the options given below:

- a. C and D b. D and A
c. A and B d. B and E
2. In a plant, black seed color (BB/Bb) is dominant over white seed color (bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it? (2024)
- a. Bb b. BB/Bb c. BB d. bb
3. A pink flowered Snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny? (2024)
- a. Only pink flowered plants
b. Red, Pink as well as white flowered plants
c. Only red flowered plants
d. Red flowered as well as pink flowered plants
4. Match List-I with List-II. (2024)

List-I		List-II	
(A)	Two or more alternative forms of a gene	(I)	Back cross
(B)	Cross of F ₁ progeny with homozygous recessive parent	(II)	Ploidy
(C)	Cross of F ₁ progeny with any of the parents	(III)	Allele
(D)	Number of chromosome sets in plant	(IV)	Test cross

Choose the correct answer from the options given below:

- a. A-(III), B-(IV), C-(I), D-(II)
b. A-(IV), B-(III), C-(III), D-(I)
c. A-(I), B-(II), C-III, D-(IV)
d. A-(II), B-(I), C-(III), D-(IV)

5. Which one of the following can be explained on the basis of Mendel's Law of Dominance? (2024)

A. Out of one pair of factors one is dominant and the other is recessive.
B. Alleles do not show any expression and both the characters appears as such in F₂ generation.
C. Factors occur in pairs in normal diploid plants.
D. The discrete unit controlling a particular character is called factor.
E. The expression of only one of the parental characters is found in a monohybrid cross.

Choose the correct answer from the options given below:

- a. B, C and D only b. A, B, C, D and E
c. A, B and C only d. A, C, D and E only
6. As per ABO blood grouping system, the blood group of father is B⁺, mother is A⁺ and child is O⁺. Their respective genotype can be (2024)

A. $I^B i / I^A i / i i$ B. $I^B I^B / I^A I^A / i i$ C. $I^A I^B / i I^A / I^B i$
D. $I^A i / I^B i / I^A i$ E. $i I^B / i I^A / I^A I^B$

Choose the most appropriate answer from the options given below:

- a. C & B only b. D & E only
c. A only d. B only
7. A heterozygous pea plant with violet flowers was crossed with homozygous pea plant with white flower. Violet is dominant over white. Which one of the following represents the expected combinations among 40 progenies formed? (2023-Manipur)
- a. 30 produced violet and 10 produced white flowers
b. 20 produced violet and 20 produced white flowers
c. All 40 produced violet flowers
d. All 40 produced white flowers

8. Given below are two statements:

Statement I: Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance.

Statement II: Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height.

In the light of the above statements, choose the correct answer from the options given below. (2022)

- a. Statement I is incorrect but Statement II is correct
b. Both Statement I and Statement II are correct
c. Both statement I and statement II are incorrect
d. Statement I is correct but Statement II is incorrect

9. The production for gametes by the parents, formation of zygotes, the F_1 and F_2 plants, can be understood from a diagram called: (2021)
 a. Punch square b. Punnett square
 c. Net square d. Bullet square
10. Identify the wrong statement with reference to the gene 'I' that controls ABO blood groups. (2020)
 a. A person will have only two of the three alleles.
 b. When I^A and I^B are present together, they express same type of sugar.
 c. Allele ' i ' does not produce any sugar
 d. The gene I has three alleles.
11. How many true breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits? (2020)
 a. 2 b. 14 c. 8 d. 4
12. The number of contrasting characters studied by Mendel for his experiments was: (2020-Covid)
 a. 4 b. 2 c. 7 d. 14
13. In *Antirrhinum* (Snapdragon), a red flower was crossed with a white flower and in F_1 generation pink flowers were obtained. When pink flowers were selfed, the F_2 generation showed white, red and pink flowers. Choose the incorrect statement from the following : (2019)
 a. This experiment does not follow the principle of dominance.
 b. Pink colour in F_1 is due to incomplete dominance.
 c. Ratio of F_2 is $\frac{1}{4}$ (Red) : $\frac{2}{4}$ (Pink) : $\frac{1}{4}$ (White)
 d. Law of segregation does not apply in this experiment
14. Which of the following characteristics represent 'Inheritance of blood groups' in humans? (2018)
 A. Dominance B. Co-dominance
 C. Multiple alleles D. Incomplete dominance
 E. Polygenic inheritance
 a. B, C and E b. A, B and C
 c. B, D and E d. A, C and E
15. Among the following characters, which one was not considered by Mendel in his experiments on pea? (2017-Delhi)
 a. Stem – Tall or Dwarf
 b. Trichomes – Glandular or non-glandular
 c. Seed – Green or Yellow
 d. Pod – Inflated or Constricted
16. The genotypes of a Husband and Wife are $I^A I^B$ and $I^A i$. Among the blood types of their children, how many different genotypes and phenotypes are possible? (2017-Delhi)
 a. 3 genotypes ; 3 phenotypes
 b. 3 genotypes ; 4 phenotypes
 c. 4 genotypes ; 3 phenotypes
 d. 4 genotypes ; 4 phenotypes
17. Which one from those given below is the period for Mendel's hybridisation experiments? (2017-Delhi)
 a. 1856 - 1863 b. 1840 - 1850
 c. 1857 - 1869 d. 1870 - 1877
18. A true breeding plant is: (2016 - II)
 a. Nearly homozygous and produces offspring of its own kind
 b. Always homozygous recessive in its genetic constitution
 c. One that is able to breed on its own
 d. Produced due to cross-pollination among unrelated plants
19. In a test cross involving F_1 dihybrid flies, more parental-type offspring were produced than the recombinant-type offspring. This indicates (2016 - I)
 a. The two genes are located on two different chromosomes
 b. Chromosomes failed to separate during meiosis
 c. The two genes are linked and present on the same chromosome
 d. Both of the characters are controlled by more than one gene
20. A tall true breeding garden pea plant is crossed with a dwarf true breeding garden pea plant. When the F_1 plants were selfed the resulting genotypes were in the ratio of: (2016 - I)
 a. 1 : 2 : 1 : Tall homozygous : Tall heterozygous : Dwarf
 b. 1 : 2 : 1 : Tall heterozygous : Tall homozygous : Dwarf
 c. 3 : 1 : Tall : Dwarf
 d. 3 : 1 : Dwarf : Tall
21. How many pairs of contrasting characters in pea plants were studied by Mendel in his experiments? (2016 - I)
 a. Eight b. Seven c. Five d. Six
22. A man with blood group 'A' marries a woman with blood group 'B'. What are all the possible blood groups of their offspring's? (2016 - I)
 a. A, B, AB and O b. O only
 c. A and B only d. A, B and AB only
23. Multiple alleles are present: (2015)
 a. At the same locus of the chromosome
 b. On non-sister chromatids
 c. On different chromosomes
 d. At different loci on the same chromosome
24. Alleles are: (2015)
 a. Different molecular forms of a gene
 b. Heterozygote's
 c. Different phenotype
 d. True breeding homozygote's
25. A gene showing co-dominance has: (2015 Re)
 a. Alleles tightly linked on the same chromosome
 b. Alleles that is recessive to each other
 c. Both alleles independently expressed in the heterozygote
 d. One allele dominant on the other
26. In his classic experiments on pea plants, Mendel did not use: (2015 Re)
 a. Pod length b. Seed shape
 c. Flower position d. Seed colour

27. If two persons with 'AB' blood group marry and have sufficiently large number of children, these children could be classified as 'A' blood group : 'AB' blood group : 'B' blood group in 1 : 2 : 1 ratio. Modern technique of protein electrophoresis reveals presence of both 'A' and 'B' type proteins in 'AB' blood group individuals. This is an example of: (2013)
- a. Complete dominance b. Co-dominance
c. Incomplete dominance d. Partial dominance
28. Which Mendelian idea is depicted by a cross in which the F_1 generation resembles both the parents? (2013)
- a. Co-dominance b. Incomplete dominance
c. Law of dominance d. Inheritance of one gene

Inheritance of Two Genes

29. When a tall pea plant with round seeds was selfed, it produced the progeny of: (2024 Re)
- A. tall plants with round seeds and
B. tall plants with wrinkled seeds.
Identify the genotype of the parent plant.
- a. TtRr b. TtRR c. TTrr d. TTRr
30. Match List-I with List-II. (2023-Manipur)

List-I (Type of cross)		List-II (Phenotypic Ratio)	
(A)	Monohybrid Cross	(I)	1 : 1
(B)	Dihybrid Cross	(II)	1 : 2 : 1
(C)	Incomplete dominance	(III)	3 : 1
(D)	Test Cross	(IV)	9 : 3 : 3 : 1

Choose the **correct** answer from the options given below:

- a. A-(III), B-(IV), C-(II), D-(I)
b. A-(II), B-(IV), C-(III), D-(I)
c. A-(II), B-(III), C-(IV), D-(I)
d. A-(IV), B-(III), C-(I), D-(II)
31. A certain plant homozygous for yellow seeds and red flowers was crossed with a plant homozygous for green seeds and white flowers. The F_1 plants had yellow seeds and pink flowers. The F_1 plants were selfed to get F_2 progeny. Assuming independent assortment of the two characters, how many phenotypic categories are expected for these characters in the F_2 generation? (2023-Manipur)
- a. 9 b. 16 c. 4 d. 6
32. What is the expected percentage of F_2 progeny with yellow and inflated pod in dihybrid cross experiment involving pea plants with green coloured, inflated pod and yellow coloured constricted pod? (2022 Re)
- a. 9% b. 100% c. 56.25% d. 18.75%
33. Fruit colour in squash is an example of: (2014)
- a. Inhibitory genes b. Recessive epistasis
c. Dominant epistasis d. Complementary genes

Chromosomal Theory of Inheritance

34. The chromosomal theory of inheritance was proposed by (2022 Re)
- a. Robert Brown b. Thomas Morgan
c. Sutton and Boveri d. Gregor Mendel
35. Experimental verification of the chromosomal theory of inheritance was done by: (2020)
- a. Sutton b. Boveri c. Morgan d. Mendel
36. Chromosomal theory of inheritance was proposed by: (2020-Covid)
- a. Bateson and Punnett b. T.H. Morgan
c. Watson and Crick d. Sutton and Boveri
37. What map unit (Centimorgan) is adopted in the construction of genetic maps? (2019)
- a. A unit of distance between two expressed genes representing 10% cross over.
b. A unit of distance between two expressed genes representing 100% cross over.
c. A unit of distance between genes on chromosomes, representing 1% cross over.
d. A unit of distance between genes on chromosomes, representing 50% cross over.

Linkage and Recombination, Polygenic Inheritance and Pleiotropy

38. Given below are two statements:
- Statement I:** When many alleles of a single gene govern a character, it is called polygenic inheritance.
Statement II: In Polygenic inheritance, the effect of each allele is additive.
- In the light of the above statements, choose the correct answer from the options given below. (2024 Re)
- a. Statement I is true but Statement II is false
b. Statement I is false but Statement II is true
c. Both Statement I and Statement II are true
d. Both Statement I and Statement II are false

39. Match List-I with List-II: (2024 Re)

	List-I Type of Inheritance		List-II Example
A.	Incomplete dominance	(I)	Blood groups in human
B.	Co-dominance	(II)	Flower colour in Antirrhinum
C.	Pleiotropy	(III)	Skin colour in human
D.	Polygenic inheritance	(IV)	Phenylketonuria

Choose the **correct** answer from the option given below:

- a. A-(III), B-(IV), C-(II), D-(I)
b. A-(II), B-(I), C-(IV), D-(III)
c. A-(II), B-(III), C-(I), D-(IV)
d. A-(IV), B-(I), C-(III), D-(II)

40. The phenomenon of pleiotropism refers to (2023)
- more than two genes affecting a single character.
 - presence of several alleles of a single gene controlling a single crossover.
 - presence of two alleles, each of the two genes controlling a single trait.
 - a single gene affecting multiple phenotypic expression.
41. Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by (2023)
- Henking
 - Thomas Hunt Morgan
 - Sutton and Boveri
 - Alfred Sturtevant
42. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
- Assertion (A):** Mendel's law of Independent assortment does not hold good for the genes that are located closely on the same chromosome.
- Reason (R):** Closely located genes assort independently.
- In the light of the above statements, choose the correct answer from the options given below. (2022)
- (A) is not correct but (R) is correct
 - Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (A) is correct but (R) is not correct
43. The best example for pleiotropy is: (2020-Covid)
- Phenylketonuria
 - Colour Blindness
 - ABO Blood group
 - Skin colour
44. The frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes was explained by (2019)
- T.H. Morgan
 - Gregor J. Mendel
 - Alfred Sturtevant
 - Sutton Boveri
45. Select the correct statement: (2018)
- Franklin Stahl coined the term "linkage".
 - Punnett square was developed by a British scientist.
 - Spliceosomes take part in translation.
 - Transduction was discovered by S. Altman.
46. Match the terms in Column I with their description in Column II and choose the correct option (2016 - I)

	Column I		Column II
A.	Dominance	(I)	Many genes govern a single character
B.	Co-dominance	(II)	In a heterozygous organism only one allele expresses itself
C.	Pleiotropy	(III)	In a heterozygous organism both alleles express themselves fully
D.	Polygenic inheritance	(IV)	A single gene influences many characters

- A-(II), B-(I), C-(IV), D-(III)
 - A-(II), B-(III), C-(IV), D-(I)
 - A-(IV), B-(I), C-(II), D-(III)
 - A-(IV), B-(III), C-(I), D-(II)
47. The movement of a gene from one linkage group to another is called: (2015)
- Translocation
 - Crossing over
 - Inversion
 - Duplication
48. The term "linkage" was coined by: (2015 Re)
- T. Boveri
 - G. Mendel
 - W. Sutton
 - T.H. Morgan
49. A pleiotropic gene: (2015 Re)
- Is a gene evolved during Pliocene
 - Controls a trait only in combination with another gene
 - Controls multiple traits in an individual
 - Is expressed only in primitive plants
50. Which of the following statements is not true for two genes that show 50% recombination frequency? (2013)
- If the genes are present on the same chromosome, they undergo more than one crossover in every meiosis
 - The genes may be on different chromosomes
 - The genes are tightly linked
 - The genes show independent assortment

Sex Determination And Mutation

51. Select the correct statement/s with respect to mechanism of sex determination in Grasshopper. (2023-Manipur)
- It is an example of female heterogamety.
 - Male produces two different types of gametes either with or without X chromosome.
 - Total number of chromosomes (autosomes and sex chromosomes) is same in both males and females.
 - All eggs bear an additional X chromosome besides the autosomes.
- Choose the **correct** answer from the options given below:
- (B) and (D) only
 - (A), (C) and (D) only
 - (A) only
 - (A) and (C) only
52. Select the incorrect statement. (2019)
- Male fruit fly is heterogametic
 - In male grasshoppers 50% of sperms have no sex-chromosome
 - In domesticated fowls, sex of progeny depends on the type of sperm rather than egg
 - Human males have one of their sex-chromosome much shorter than the other
53. Which of the following pairs is wrongly matched? (2018)
- Starch synthesis in pea : Multiple alleles
 - ABO blood grouping : Co-dominance
 - XO type sex determination : Grasshopper
 - T.H. Morgan : Linkage

54. An abnormal human baby with 'XXX' sex chromosomes was born due to: (2015)
- Fusion of two ova and one sperm
 - Fusion of two sperms and one ovum
 - Formation of abnormal sperms in the father
 - Formation of abnormal ova in the mother

Genetic Disorders

55. Given below are two statements: (2024 Re)

Statement I: Failure of segregation of chromatids, during cell cycle resulting in the gain or loss of whole set of chromosome in an organism is known as aneuploidy.

Statement II: Failure of cytokinesis after anaphase stage of cell division results in the gain or loss of a chromosome is called polyploidy.

In the light of the above statements, choose the correct answer from the options given below:

- Statement I is true but Statement II is false
 - Statement I is false but Statement II is true
 - Both Statement I and Statement II are true
 - Both Statement I and Statement II are false
56. Aneuploidy is a chromosomal disorder where chromosome number is not the exact copy of its haploid set of chromosomes, due to: (2024 Re)
- Substitution
 - Addition
 - Deletion
 - Translocation
 - Inversion

Choose the most appropriate answer from the options given below:

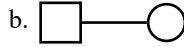
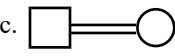
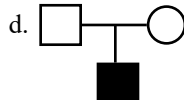
- C and D only
 - D and E only
 - A and B only
 - B and C only
57. Match List-I with List-II: (2024)

List-I		List-II	
(A)	Down's syndrome	(I)	11 th chromosome
(B)	α -Thalassemia	(II)	'X' chromosome
(C)	β -Thalassemia	(III)	21 st chromosome
(D)	Klinefelter's syndrome	(IV)	16 th chromosome

Choose the correct answer from the options given below:

- A-(III), B-(IV), C-(I), D-(II)
 - A-(IV), B-(I), C-(II), D-(III)
 - A-(I), B-(II), C-(III), D-(IV)
 - A-(II), B-(III), C-(IV), D-(I)
58. Which of the following statements are correct about Klinefelter's Syndrome? (2023)
- This disorder was first described by Langdon Down (1866).
 - Such an individual has overall masculine development. However, the feminine development is also expressed.
 - The affected individual is short stature.
 - Physical, psychomotor and mental development is retarded.
 - Such individuals are sterile.

Choose the correct answer from the options given below:

- A and E only
 - A and B only
 - C and D only
 - B and E only
59. Which one of the following symbols represents mating between relatives in human pedigree analysis? (2023)
- 
 - 
 - 
 - 

60. Broad palm with single palm crease is visible in a person suffering from- (2023)

- Thalassemia
 - Down's syndrome
 - Turner's syndrome
 - Klinefelter's syndrome
61. Given below are two statements: (2022 Re)

Statement-I: Sickle cell anaemia and Haemophilia are autosomal dominant traits.

Statement-II: Sickle cell anaemia and Haemophilia are disorders of the blood.

In the light of the above statements, choose the correct answer from the options given below:

- Statement-I is incorrect but Statement-II is correct
 - Both Statement-I and Statement-II are correct
 - Both Statement-I and Statement-II are incorrect
 - Statement-I is correct but Statement-II is incorrect
62. If a female individual is with small round head, furrowed tongue, partially open mouth and broad palm with characteristic palm crease. Also the physical, psychomotor and mental development is retarded. The karyotype analysis of such an individual will show (2022 Re)
- Trisomy of chromosome 21
 - 47 chromosomes with XXY sex chromosomes
 - 45 chromosomes with XO sex chromosomes
 - 47 Chromosomes with XYY sex chromosomes

63. Select the incorrect match regarding the symbols used in Pedigree analysis (2022 Re)

-  Parent with male child affected with disease
-  Sex unspecified
-  Affected individual
-  Consanguineous mating

64. A normal girl, whose mother is haemophilic marries a male with no ancestral history of haemophilia. What will be the possible phenotypes of the offsprings? (2022 Re)

- Haemophilic son and haemophilic daughter.
- Haemophilic son and carrier daughter.
- Normal daughter and normal son.
- Normal son and haemophilic daughter.

Choose the most appropriate answer from the options given below:

- (B) and (D) only
- (A) and (B) only
- (B) and (C) only
- (A) and (D) only

65. Which of the following occurs due to the presence of autosome linked dominant trait? (2022)

- a. Thalassemia b. Sick cell anaemia
c. Myotonic dystrophy d. Haemophilia

66. If a colour blind female marries a man whose mother was also colour blind, what are the chances of her progeny having colour blindness? (2022)

- a. 100% b. 25% c. 50% d. 75%

67. In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased? (2021)

- a. 75% b. 25% c. 100% d. 50%

68. Select the correct match (2020)

	Column - I		Column - II
a.	Phenylketonuria	-	Autosomal dominant trait
b.	Sickle cell anaemia	-	Autosomal recessive trait, chromosome-11
c.	Thalassemia	-	X linked
d.	Haemophilia	-	Y linked

69. What is the genetic disorder in which an individual has an overall masculine development gynaecomastia, and is sterile? (2019)

- a. Turner's syndrome b. Klinefelter's syndrome
c. Edward syndrome d. Down's syndrome

70. A woman has an X-linked condition on one of her X chromosomes. This chromosome can be inherited by: (2018)

- a. Only daughters b. Only sons
c. Only grandchildren d. Both sons and daughters

71. A disease caused by an autosomal primary non-disjunction is (2017-Delhi)

- a. Down's syndrome b. Klinefelter's syndrome
c. Turner's syndrome d. Sick cell anemia

72. Thalassemia and sickle cell anemia are caused due to a problem in globin molecule synthesis. Select the correct statement. (2017-Delhi)

- a. Both are due to a qualitative defect in globin chain synthesis
b. Both are due to a quantitative defect in globin chain synthesis
c. Thalassemia is due to less synthesis of globin molecules
d. Sick cell anemia is due to a quantitative problem of globin molecules

73. If a colour-blind man marries a woman who is homozygous for normal colour vision, the probability of their son being colour-blind is: (2016 - II)

- a. 0.75 b. 1 c. 0 d. 0.5

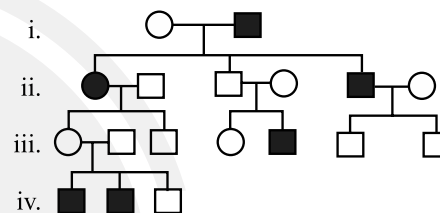
74. Pick out the correct statements: (2016 - I)

- A. Haemophilia is a sex-linked recessive disease.
B. Down's syndrome is due to aneuploidy.
C. Phenylketonuria is an autosomal recessive gene disorder.
D. Sick cell anaemia is an X-linked recessive gene disorder.
a. (A) and (D) are correct
b. (B) and (D) are correct
c. (A), (C) and (D) are correct
d. (A), (B) and (C) are correct

75. Which of the following most appropriately describes haemophilia? (2016 - I)

- a. Recessive gene disorder
b. X-linked recessive gene disorder
c. Chromosomal disorder
d. Dominant gene disorder

76. In the following human pedigree, the filled symbols represent the affected individuals. Identify the type of given pedigree. (2015 Re)



- a. X-linked recessive b. Autosomal recessive
c. X-linked dominant d. Autosomal dominant

77. A colour blind man marries a woman with normal sight who has no history of colour blindness in her family. What is the probability of their grandson being colour blind? (2015 Re)

- a. 1 b. Nil c. 0.25 d. 0.5

78. A human female with Turner's syndrome: (2014)

- a. Is able to produce children with normal husband
b. Has 45 chromosomes with XO
c. Has one additional X chromosome
d. Exhibits male characters

79. A man whose father was colour blind marries a woman who had a colour blind mother and normal father. What percentage of male children of this couple will be colour blind? (2014)

- a. 75% b. 25% c. 0% d. 50%

80. The incorrect statement with regard to Haemophilia is: (2013)

- a. A single protein involved in the clotting of blood is affected
b. It is a sex - linked disease
c. It is a recessive disease
d. It is a dominant disease

81. If both parents are carriers for Thalassaemia, which is an autosomal recessive disorder, what are the chances of pregnancy resulting in an affected child? (2013)

- a. 100% b. No chance c. 50% d. 25%

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (d) | 3. (d) | 4. (a) | 5. (d) | 6. (c) | 7. (b) | 8. (b) | 9. (b) | 10. (b) |
| 11. (b) | 12. (c) | 13. (d) | 14. (b) | 15. (b) | 16. (c) | 17. (a) | 18. (a) | 19. (c) | 20. (a) |
| 21. (b) | 22. (a) | 23. (a) | 24. (a) | 25. (c) | 26. (a) | 27. (b) | 28. (a) | 29. (d) | 30. (a) |
| 31. (d) | 32. (d) | 33. (c) | 34. (c) | 35. (c) | 36. (d) | 37. (c) | 38. (b) | 39. (b) | 40. (d) |
| 41. (d) | 42. (d) | 43. (a) | 44. (c) | 45. (b) | 46. (b) | 47. (a) | 48. (d) | 49. (c) | 50. (c) |
| 51. (a) | 52. (c) | 53. (a) | 54. (d) | 55. (d) | 56. (d) | 57. (a) | 58. (d) | 59. (c) | 60. (b) |
| 61. (a) | 62. (a) | 63. (d) | 64. (c) | 65. (c) | 66. (a) | 67. (b) | 68. (b) | 69. (b) | 70. (d) |
| 71. (a) | 72. (c) | 73. (c) | 74. (d) | 75. (b) | 76. (b) | 77. (c) | 78. (b) | 79. (d) | 80. (d) |
| 81. (d) | | | | | | | | | |

